

MedCare4Home: A Mobile Medical System for Families

¹Mengting Wang¹Yen Le²Chen-Hsiang Yu

¹School of Computing and Data Science
Wentworth Institute of Technology
Boston, MA, U.S.A.
{wangm1, ley}@wit.edu

²Multidisciplinary Graduate Engineering
College of Engineering, Northeastern University
Boston, MA, U.S.A.
jones.yu@northeastern.edu

Abstract - Managing healthcare data is essential for everyone, but it can be challenging and time-consuming. Although several Web and mobile applications existed for personal medical record management, there were few home-based, family-oriented solutions. In this research, we proposed to design an easy-to-use solution for a family. We designed and implemented an innovative Home Medical Care System, *MedCare4Home*, by deploying a MERN stack Web application to a minicomputer, Raspberry Pi 4, and enabling a local network to connect different devices at home. The users can access the Web application using either a desktop, tablet, mobile phone, or simply a 7-inch LCD touchscreen. *MedCare4Home* not only allows users to keep track of appointments and organize healthcare documents, but it can also help set up medication reminders and provide a self-report interface for symptom collection for family members easily and securely in one place. The final developed system demonstrated that we achieved the original goal and beyond. We expect to extend our current hardware design and software development to support physiological data collection with external sensors in the future.

Keywords - Healthcare, data management, Web application, Internet of Things (IoT)

I. INTRODUCTION

The healthcare industry is relatively slow to join the global digital revolution, but new technologies present a good promise to transform health services. These technologies allow doctors, hospitals and health systems to do more things, such as filtering, sorting and organizing quickly and accurately the massive amounts of information [25].

On the other hand, the need to digitize healthcare data is increasing [1] and managing healthcare data is essential to everyone. Virtual healthcare allows patients to easily access primary care through a video interface on computers or mobile devices and there was an urgent need for virtual healthcare to reduce the risk of community spread and protect patients [2]. It is crucial to properly manage health data and retrieve the data easily. However, Health Data Management (HDM) is tasked not only with organizing medical data from different medical sources, but also integrates the analysis to make personal medical care more efficient. HDM can get more challenging when you need to care for minors or other loved ones in your family.

People may also want to perform some basic health checks at home to keep track of health for themselves or family members, monitor ongoing conditions, or check symptoms. For example, one of the simple home medical tests is the blood pressure test

[3]. According to the CDC, high blood pressure is a common disease in the United States; tens of millions of adults have high blood pressure, and many do not have it under control. High blood pressure increases the risk of heart disease and stroke, which are two leading causes of death for Americans, and often there are no warning symptoms [4]. Hence, regularly checking blood pressure and recording home test results is vital.

Currently, most of the existing mobile or web applications for personal medical records focus on medical information collected from different patient portals, and few are family-oriented. Even though some applications offer managing health records for family members, they might lack support for external health monitoring devices, and the home medical tests only support manual inputs from users.

To support family health data management, we propose a new system, *MedCare4Home*, which is an easy-to-use web application deployed on a minicomputer and provides local access to other devices. We performed frequent user tests throughout the design and development phases to maximize the usability of the application. Apart from organizing healthcare documents and recording home medical test results, some key designs include setting up profiles for family members, managing appointments and prescriptions, and supporting self-reporting symptoms. Our system stores all user data securely on a local MongoDB database running on the minicomputer to protect sensitive health data.

In the following paper, we will talk about related work, system design and implementation. Conclusions, discussion and future work will be presented at the end.

II. RELATED WORK

Currently, many hospitals offer online portals to allow patients to access their health information. For example, Patient Gateway, Mass General Hospital's digital patient portal, is a well-developed website to help manage patient's health and communication with doctor's offices efficiently, and securely [21]. In addition to displaying all records, patients can also schedule an in-person or virtual appointment conveniently. The website offers multiple ways to share health information, and the patients can grant full access to friends and family, give one-time limited access to any person or website, or link accounts with other healthcare organizations to manage health data in one place. However, since not all health organizations are connected, users might still have to log in to different websites

or use a combination of paper and electrical methods to manage all important health data. Similarly, other hospitals, such as Tufts Medical Center, Boston's Children's Hospital, Mayo Clinic, and so on, provide the same services to help patients with medical care and health document management. For people who are not tech-savvy, it can be overwhelming and challenging to navigate through a website with rich features.

There are some popular Personal Health Record (PHR) web or mobile applications available in the market that consolidate health records from different resources, such as HealthVault, MTBC PHR, myPHR [20], and so on, but few of them are home-based and family-oriented. The merging of healthcare and technology for users is critical; even giant companies like Apple have innovated their own software to help organize important personal health information.

As mentioned on Apple's official website, "It [the Health app] collects data from your iPhone, the built-in sensors on your Apple Watch, compatible medical devices, and apps that use HealthKit." According to Apple, they will add several new features to the Health app when they release newer iOS versions [5]. Users will be able to share their Health app data with loved ones and provide useful information to healthcare providers. This feature will begin with support from six electronic medical records (EMR) companies in the United States, such as Cerner and Athenahealth. It means doctors using these platforms can quickly access patients' additional medical data efficiently [6]. Although the Health app is convenient and easy to use, it is only available for Apple Smart device owners, which leaves numerous non-Apple users out of Apple Health. OneRecord is a solution to avoid the cross-platform availability issue as users can access it through the website or download the app from Apple Store or Google Play Store. Unlike Apple, the OneRecord app is more specialized in healthcare [7]. However, the OneRecord app lacks features to record home tests or report symptoms because its primary focuses are health record management and family sharing.

III. PROPOSED IDEA AND HYPOTHESIS

We proposed to have a home medical care system to help families easily manage healthcare data at home. With our solution, users can seamlessly manage their medical documents, prescriptions, and appointments in a central place at home. Our proposed idea focuses on families rather than individuals. It supports multiple profiles for other members within one account to allow users to manage the whole family in one place. In addition, we provide features to self-report ongoing symptoms and record home medical test results, which are valuable information for the next doctor's appointment.

We have two hypotheses for the *MedCare4Home* system. Firstly, our proposed system is easy to set up at home and can be carried easily since it is pocket-sized and lightweight. For example, the users can bring the system to a doctor's appointment to present health data or monitor family health conditions even when they are traveling. Secondly, the system should also work within the local network for users without reliable Internet access. According to the Federal Communications Commission (FCC), approximately six percent of the population still lacks access to fixed broadband service [8]. We expect all families to take advantage of the proposed home medical system.

IV. SYSTEM DESIGN

The proposed system, *MedCare4Home*, is a web application deployed in a minicomputer running on a local network that allows users to use any mobile device to access it. Although there are diverse web technologies and hardware options, the design considered using the MERN stack for software development and Raspberry Pi 4 as a minicomputer for application hosting and deployment.

4.1 Paper Prototyping and Prototype Testing

The design of the proposed system follows Human-Computer Interaction (HCI) principles and has gone through user-centered design. The paper prototypes were created (Figure 1) and tested with ten users. In summary, most users respond that "simple is the best" and they prefer a clean user interface. In addition, users recommended using neutral colors, such as blue or grey, because these colors are calmer and more relaxing. All users preferred to have a responsive navigation bar on the left instead of on the top. Based on the feedback and suggestions, we combined content from separate pages to reduce users' work and made icons more prominent for users to locate quickly. In addition, we highlighted upcoming appointments, medication, and the required medical test on the home page. To support navigation, we listed all the features on the navigation bar for quick access and the users can switch profiles quickly by clicking on the settings icon on the top right.

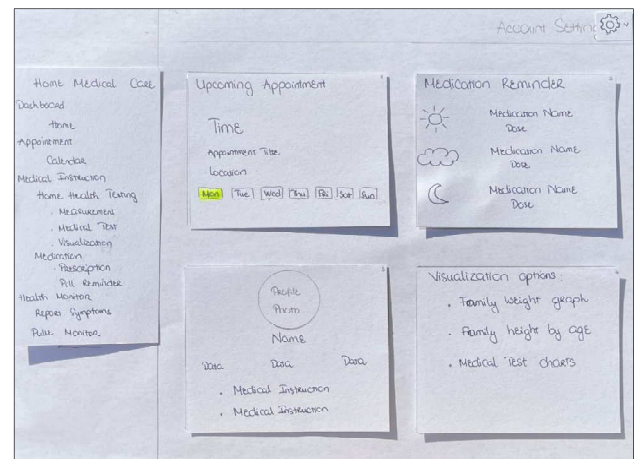


Figure 1. Paper prototyping of the Web application.

4.2 Software Architecture

We decided to build a Web-based application using the MERN stack. The MERN stack stands for MongoDB, Express.js, React and Node.js. It is a collection of robust and powerful technologies. It uses JavaScript as the primary programming language throughout the stack, which is efficient and helps developers follow a modern approach to web development [23]. While React is the JavaScript frontend library, Express.js is a fast server-side application framework to support Node.js, a Cross-platform JavaScript framework, and MongoDB is a modern database system. Advantages of using the MERN stack include Model View Controller (MVC) architecture for a smooth process of development, an open-source framework for end-to-end global community support [9]. The software architecture of the Web-based application is illustrated as Figure 2.

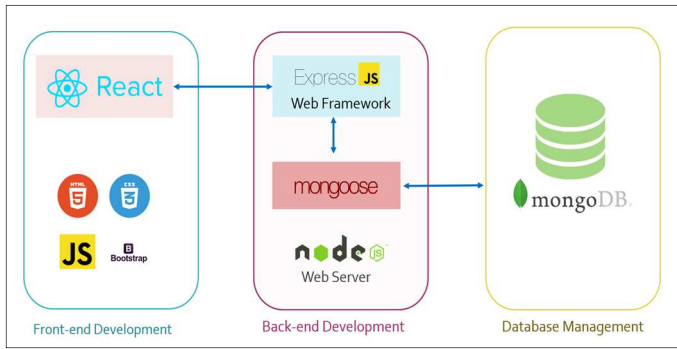


Figure 2. Software architecture of the Web-based application

4.3 Data Management

Another key factor we considered during the decision-making process for the development stack was to understand whether relational or non-relational database architecture is better for our application. NoSQL databases offer many benefits over relational databases, such as more flexible data models, horizontal scalabilities, fast queries, and ease for developers to work with [10]. Unlike relational databases, MongoDB stores its data in structured documents rather than the fixed tables. The first step of the design was to design document models.

4.4 Hardware Architecture

To allow easily use the proposed system in home setting, we not only deployed the built Web-based application to a minicomputer, i.e., Raspberry pi 4, but we also provided the user interface via either 7-inch LCD or any devices connected to the local network via a router. (Figure 4) The Raspberry Pi is a popular low-cost, energy-efficient, small-sized, expandable minicomputer that is powerful enough to build the proposed system. Although Raspberry Pi 3 B+ or Raspberry Pi Zero are capable to run our application and the price can go as low as \$10, we eventually decided to use the Raspberry Pi 4 B (8GB) version because it has more advantages, including improvement in processor speed, multimedia performance, memory, and connectivity [11]. Another hardware piece of the proposed system is a router that enables Internet or Intranet connection for users.

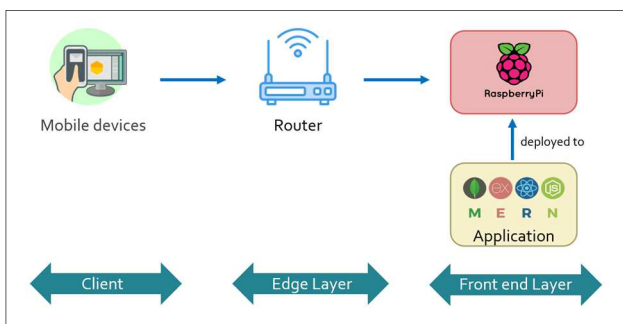


Figure 4. Hardware architecture of the whole system

V. IMPLEMENTATION

A. Front-end Development

The *MedCare4Home* system is proposed to help users manage their medical data for family members. They can access the data from anywhere at any time. To quickly switch between multiple profiles, we needed a flexible Web stack that can update data

without page reloading. We used React to create reusable UI components to present data that changes over time. In traditional JavaScript application, we need to look at what data changes and make such changes to the. Instead, React takes a different approach. When the component is initialized, the render method is called to generate a lightweight representation of the view. When the data is changed, the render method is called again. To perform updates, we differentiate the return value from the previous call to render with the new one and generate a minimal set of changes to the DOM [12].

One challenge of the Web-based application was data passing between each component by states. We used Redux as the state management tool to manage the application state in a single place, facilitated communication and shared data across components [13]. Redux works as the following. There is a central store that holds the entire state of the application, and each component can access the stored state without having to send down props from one component to another. There are three building parts: action, store, and reducers. We can consider actions as events. They are the way we can send data from user interactions, API calls, or form submissions. Actions are sent using the "store.dispatch()" method and it can have a payload of the information for the action. Reducers are functions that take the current state of an application, perform an action, and return a new state. These states are stored as objects, and they specify how the state of an application changes in response to an action sent to the store. The store holds the application state. It is highly recommended to keep only one store in any Redux application [13].

B. Back-end Development

On the backend, we used MongoDB to store user information. It is a document-based database that stores data in JSON-like documents; these documents also support arrays and nested objects. As stated, "data in MongoDB has a flexible schema. Collections do not enforce document structure by default" [14]. This means developers can choose a data model based on application and performance requirements accordingly. We handled one-to-many relationships through Embedding rather than References. Document Reference is more similar to associations in relational databases than embedded. However, the embedded-document structure allowed us to quickly retrieve and manipulate related data in a single database operation [4]. With only the users and profiles documents in the collection, we could get any data for each profile for a family member efficiently.

To work with MongoDB conveniently and efficiently, we used Mongoose, a Node.js-based Object Data Modeling (ODM) library for MongoDB [15, 16]. Object Relational Mapper (ORM), such as SQLAlchemy, maps objects to records in relational database tables. Similarly, ODM maps objects to documents in MongoDB. Mongoose can enforce a specific database schema at the application layer while allowing for great flexibility when needed.

As Figure 4 illustrates, we set up Node.js in a minicomputer (Raspberry Pi 4) to build the lightweight Web server in the local network. We then deployed the MERN stack-based web application to it. There are two ways to access the Web-based

application deployed in the minicomputer (Raspberry Pi 4): (1) the user can use any personal device, such as smartphones or tablets, via the router, or (2) directly interact with a 7-inch LCD touchscreen display connected to the minicomputer.

C. Technical Challenges & Solutions

Learning and working with Raspberry Pi was an incredible experience. We were amazed by how much this low-cost minicomputer can do. However, we were unaware that this device can be fragile and accidentally be while connecting to an LCD. It is crucial that we manipulate the new Raspberry Pi with extreme caution to avoid breaking the Raspberry Pi or spending a long time finding the cause of the device not powering on. Another takeaway from our experience with the Raspberry Pi was that different models of Raspberry Pi can support your needs for your application. The 2GB Raspberry Pi 3 works for our application, but it was extremely slow when installing dependencies for the front end.

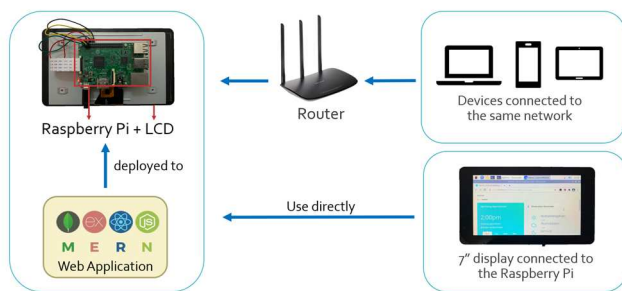


Figure 4. Hardware configuration of *MedCare4Home* system

Additionally, setting up MongoDB was one of the most difficult tasks in the process. Our Raspberry Pi 3 runs 32-bit Raspbian with a default MongoDB version 2.4.14. However, newer ARM builds of MongoDB require ARM64, but Raspbian is only 32-bit [17]. Our attempt to compile MongoDB 3.2 from the source code to meet the requirements for our application failed. We resolved the database issue by installing 64-bit Raspbian on an 8GB Raspberry 4, which also made the rest of our deployment smoother.

Furthermore, integration of the front-end and back-end was the most challenging and time-consuming. One of the most frequently encountered issues was receiving back-end triggered updates. We had to constantly check the server to ensure changes were made as expected.

VI. EVALUATION

We evaluated our application by comparing it with existing solutions from seven different dimensions: customizable, family-oriented, usability, scalability, data-security, open-source, and cross-platform support. As Table 1 shows, the *MedCare4Home* is a low-cost solution for family healthcare management at home. Based on the RAM choice, the Raspberry Pi 4 price can go as low as \$35 and users can purchase a 7-inch official display for Raspberry Pi at about \$70; the entire setup cost will add up to about \$100, which is fairly cost-effective.

Table 1: The comparison of *MedCare4Home* with other solutions

Dimensions	MedCare4Home	Apple Health app	MTBC PHR	myPHR	OneRecord	HealthVault (discontinued)
Customizable	Yes	Yes	Yes	Yes	Yes	Yes
Family-oriented	Yes	Yes	No	No	Yes	No
Usability (0-10)	9	8	5	5	8	5
Scalability (0-10)	10	7	0	0	0	0
Data-security (0-10)	9	8	7	7	0	8
Open-source	Yes	No	No	No	No	No
Cross-platform support	Yes	No (iOS only)	Yes	Yes	Yes	No (web only)

Overall, *MedCare4Home* is not only a secure and user-friendly solution for family healthcare management, but it also has great advantages in customizability and scalability. Apple's Health app is certainly a well-developed health application with rich features [5]. However, it is only available on iOS which has locked out many non-Apple users. Also, the average cost of an iPhone is significantly higher than a Raspberry Pi.

Instead, our system can be set up easily and affordable for them. Once the system is up and running, the users will be able to access all the features intuitively. In addition, the *MedCare4Home* is extremely flexible and scalable to meet different users' needs by allowing connections to various external health monitoring devices, such as smartphones, tablets, and laptops. When having personal health documents stored elsewhere, privacy and security are another concern [22]. Since all the valuable healthcare data are stored locally on the Raspberry Pi, only users themselves can access the data. Our local setup ensures users have complete control over their data, which protects user's privacy and significantly improves the user's health data security. Unlike most commercial solutions, *MedCare4Home* is open-source and available on Github for anyone to use or modify for any purpose at no cost [26].

The final implementation of *MedCare4Home* system is illustrated in Figure 5-6.

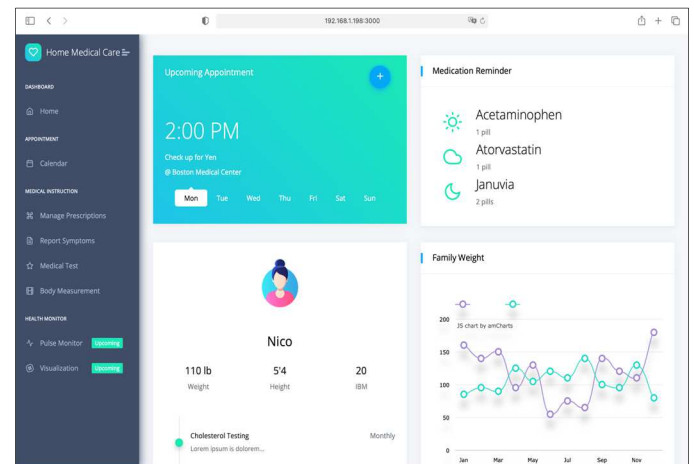


Figure 5. The user interface of Web-based application of *MedCare4Home* system

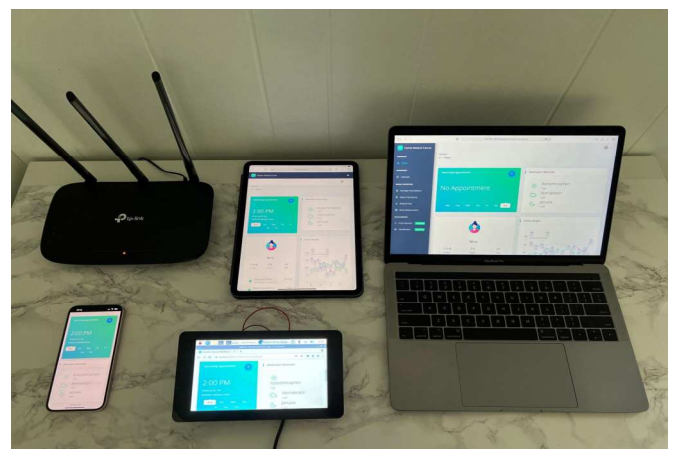


Figure 6. The hardware design of *MedCare4Home* system.

VII. DISCUSSION

Although we have successfully implemented the *MedCare4Home* system based on our design, there were some challenges we faced during the whole process we would like to discuss.

A. Platform Selection

The *MedCare4Home* was initially planned to deploy in a cloud environment, such as Amazon Web Services and Google Cloud Platform, using various cloud services. Considering users with limited Internet access in certain countries or regions, we decided to construct the application relying on a local network. This change in design allowed us to explore different hardware components. Even though the current setup provides a solution, we lost an opportunity to take advantage of key cloud computing benefits, such as high availability, rapid scalability, security, and data recovery.

B. Computational Need for Medical Usage

From our experience, although using a Raspberry Pi 3 B can minimize the cost, it's not the best fit for web hosting due to low computing power. Compared with the Raspberry Pi 3, the Raspberry Pi 4 provides upgraded specifications and significantly improves the processing speed and performance of our system. Building a Raspberry Pi RAID (Redundant Array of Inexpensive Disks) NAS Server with multiple Raspberry Pi could potentially be an option for further speed and reliability improvement. RAID is a way of logically putting multiple disks together into a single array [18]. With the right number of Raspberry Pi, we can achieve a good balance between redundancy and speed. On the other hand, although the Raspberry Pi 4 is an incredible minicomputer, it is challenging to perform machine learning training and testing on such a platform. Considering the future needs of predicting trends based on physiological data collected from external sensors, mobile AI platforms, such as Nvidia Jetson Nano or Google Coral Dev Board, might be more suitable to prototype the system than using the Raspberry Pi 4 [19].

C. Privacy issue

The key security concern for any healthcare application is user privacy [24]. Health and medical information are some of the most obvious types of information that need privacy protection. The design of *MedCare4Home* allowed the users to share relevant information with family members or others. Therefore, it needs to balance between maintaining privacy and keeping accessibility for family members. The HIPAA Privacy Rule specifies how covered entities are allowed to be disclosed. To enhance privacy, applying HIPAA privacy rules to the home medical care system is needed.

VIII. CONCLUSIONS AND FUTURE WORK

In this research, we hypothesized that users can seamlessly manage their medical documents, prescriptions, and appointments in a central place at home. We proposed *MedCare4Home* that is a home medical care system with some advantages, including cost efficient, easy to deploy and set up (at home), and convenient to carry around. In this paper, we have successfully demonstrated that it is possible to use existing

technologies, such as Web development, programming and Internet of Things (IoT), to create the proposed system. The system mainly contains two parts: hardware setup and software development. In terms of the hardware, it uses a minicomputer (Raspberry Pi), a Wi-Fi router and mobile devices, such as smartphones, tablets, laptops, etc. In addition, adding a 7-inch LCD display with the minicomputer can support easy access for the family members. The whole system setup can be run in a local network, i.e., an Intranet environment, without Internet access. For the software part, we were able to use the MERN stack (MongoDB, Express.js, React and Node.js) to create an accessible, easy-to-use Web application for the purpose. To allow more people to join the efforts, we also open-source the project to the community [26]. *MedCare4Home* is not only open source, affordable, lightweight, and easy to set up, but it also has several advantages over other existing solutions, organized in Table 1. In the future, there are a few directions to continue this research. Firstly, we want to extend our current hardware design and software development to support physiological data collection with external sensors. Secondly, we will enhance the security and privacy parts of the system. Last, but not least, we will collaborate with primary doctors and volunteer families to get objective and subjective feedback from both sides.

REFERENCES

- [1] "Health data management: Benefits, challenges and storage," *Cloudian*, 15-Feb-2021. [Online]. Available: <https://cloudian.com/guides/hipaa-compliant-cloud-storage/health-data-management/>. [Accessed: July 31, 2023].
- [2] "States' actions to expand telemedicine access during COVID-19 and future policy considerations," *States' Actions to Expand Telemedicine Access During COVID-19 | Commonwealth Fund*. [Online]. Available: <https://www.commonwealthfund.org/publications/issue-briefs/2021/jun/states-actions-expand-telemedicine-access-covid-19>. [Accessed: July 31, 2023].
- [3] "10 health checks you can do at home," *Aetna International*. [Online]. Available: <https://www.aetnainternational.com/en/about-us/explore/health-care/lifestyle-better-health/10-health-checks-you-can-do-at-home.html>. [Accessed: July 31, 2023].
- [4] "High blood pressure," *Centers for Disease Control and Prevention*, 22-Oct-2020. [Online]. Available: <https://www.cdc.gov/bloodpressure/index.htm>. [Accessed: July 31, 2023].
- [5] "iOS - Health," *Apple*. [Online]. Available: <https://www.apple.com/ios/health/>. [Accessed: July 31, 2023].
- [6] N. Wetsman, "Apple lets users see family members' Health Data," *The Verge*, 07-Jun-2021. [Online]. Available: <https://www.theverge.com/2021/6/7/22522872/health-app-family-mobility-fall-risk-wwdc-2021>. [Accessed: July 31, 2023].
- [7] J. Blumenthal, "How is OneRecord different than Apple Health?," *OneRecord*. [Online]. Available: <https://onerecord.com/blog/how-is-onerecord-different-than-apple-health>. [Accessed: July 31, 2023].
- [8] "Eighth Broadband Progress Report," *Federal Communications Commission*, 23-Aug-2012. [Online]. Available: <https://www.fcc.gov/reports-research/reports/broadband-progress-reports/eighth-broadband-progress-report>. [Accessed: July 31, 2023].
- [9] Rahul Mathur, "Mean vs mern vs MEVN vs lamp stack for development," *Technology Insights By Arka Software*, 01-Dec-2021. [Online]. Available: <https://www.arkasoftwares.com/blog/mean-vs-mern-vs-mevn-vs-lamp-stack-for-development/>. [Accessed: July 31, 2023].
- [10] "NoSQL vs SQL databases," *MongoDB*. [Online]. Available: <https://www.mongodb.com/nosql-explained/nosql-vs-sql>. [Accessed: July 31, 2023].

- [11] "Buy A raspberry pi 4 model B," *Raspberry Pi*. [Online]. Available: <https://www.raspberrypi.com/products/raspberry-pi-4-model-b>. [Accessed: July 31, 2023].
- [12] K. Gupta, "Why is mern stack our preferred platform for startups apps?," *Classic Informatics*. [Online]. Available: <https://www.classicinformatics.com/blog/why-is-mern-stack-our-preferred-platform-for-startups-apps>. [Accessed: July 31, 2023].
- [13] N. Ighodaro, "Why use redux? A tutorial with examples," *LogRocket Blog*, 14-Jun-2021. [Online]. Available: <https://blog.logrocket.com/why-use-redux-reasons-with-clear-examples-d21bffd5835>. [Accessed: July 31, 2023].
- [14] "Data Models - MongoDB Manual". [Online]. Available: <https://docs.mongodb.com/manual/data-modeling>. [Accessed: July 31, 2023].
- [15] A. Kukic, "MongoDB & mongoose: Compatibility and comparison," *MongoDB*, 25-Nov-2021. [Online]. Available: <https://www.mongodb.com/developer/article/mongoose-versus-nodejs-driver>. [Accessed: July 31, 2023].
- [16] "Data model design," *Data Model Design - MongoDB Manual*. [Online]. Available: <https://docs.mongodb.com/manual/core/data-model-design/>. [Accessed: July 31, 2023].
- [17] Emmet, "Installing mongodb on the raspberry pi," *Pi My Life Up*, 25-Nov-2021. [Online]. Available: <https://pimylifeup.com/mongodb-raspberry-pi>. [Accessed: July 31, 2023].
- [18] "(almost) everything you need to know about raid," *Steadfast*. [Online]. Available: <https://www.steadfast.net/blog/almost-everything-you-need-know-about-raid>. [Accessed: July 31, 2023].
- [19] "Dev Board," *Coral*. [Online]. Available: <https://coral.ai/products/dev-board>. [Accessed: July 31, 2023].
- [20] "Carry medical history in your pocket: Top 10 mobile apps for personal medical records," *Dr. Hempel Digital Health Network*, 28-Jul-2018. [Online]. Available: <https://www.dr-hempel-network.com/digital-health-technology/top-10-mobile-apps-for-personal-medical-records/>. [Accessed: July 31, 2023].
- [21] "Mass general Brigham Patient gateway," *Mass General Brigham*. [Online]. Available: <https://www.massgeneralbrigham.org/patient-information/what-is-patient-gateway>. [Accessed: July 31, 2023].
- [22] A. Baker, L. Vega, T. DeHart, and S. Harrison. "Medical record privacy: is it a facade?" In CHI '11 Extended Abstracts on Human Factors in Computing Systems (CHI EA '11). ACM, New York, NY, USA, 2203–2208, 2011.
- [23] T. Jungnickel, J. Cabello, and K. Raile. "HotPi: Open-Source Collaborative Patient Documentation." In *Companion of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing (CSCW '17 Companion)*. Association for Computing Machinery, New York, NY, USA, 219–222, 2017.
- [24] H. Jita and V. Pieterse. "A Framework to Apply the Internet of Things for Medical Care in a Home Environment." In *Proceedings of the 2018 International Conference on Cloud Computing and Internet of Things (CCIOT 2018)*, Association for Computing Machinery, New York, NY, USA, 45–54, 2018.
- [25] S. Edelmann, "Digitization: Future of Healthcare," *HEALTHCARE TRANSFORMERS*, 25-Feb-2021. [Online]. Available: <https://healthcaretransformers.com/healthcare-business/digitization-future-of-healthcare/>. [Accessed: July 31, 2023].
- [26] *MedCare4Home* GitHub repository. [Online] Available: <https://github.com/nicomwang/MedCare4Home> [Accessed: July 31, 2023].